Institute of Technology of Cambodia
Faculty of Electrical Engineering
Génie Informatique et Communication

GICI5CC · CLOUD COMPUTING · WEEK 1

Introduction to Cloud Computing

Definition, characteristics, history, benefits and challenges.



Heng Rathpisey · Department GIC · Academic year 2025–2026

Course overview

Institute of Technology of Cambodia · Faculty of Electrical Engineering · Génie Informatique et Communication · Academic year 2025–2026

PROGRAMME

Engineering Degree in Computer Sciences

COURSE

Cloud Computing · GICI5CC

CREDITS AND TYPE

2.5 credits · Core course · 1st semester

PREREQUISITES

Operating Systems, Systems and Networks Administration, DevOps

INSTRUCTOR

Mr. HENG Rathpisey · M.Eng. UGM Indonesia (2020) · AWS Certified Cloud Practitioner

CONTACT

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Course description

A comprehensive introduction to cloud computing principles and practice, centered on *Amazon Web Services (AWS)* and software development in the cloud.

Deployment models	Compute and storage	Networking	Databases
Cloud-native apps	Microservices	DevOps	Infrastructure as Code
Security	Cost optimization	Serverless	AI/ML services

Through lectures, laboratory exercises and a capstone project, students learn to architect cloud solutions, implement DevOps workflows and use cloud services for enterprise-grade software development.

Course learning outcomes

At the end of the course, students are able to:

CLO1 · KNOWLEDGE · C4

Analyze

cloud computing architectures, service models, AWS services and cloud-native design patterns to evaluate and recommend appropriate cloud solutions for diverse application requirements.

CLO2 · SKILL · P5

Implement

scalable cloud applications using AWS services, DevOps practices, Infrastructure as Code and containerization technologies to deploy production-ready solutions.

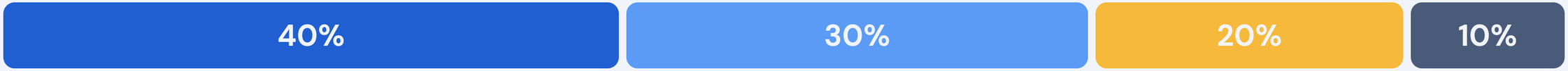
CLO3 · ATTITUDE · A4

Document

cloud architectures, configurations and implementation decisions through professional lab reports and project documentation, demonstrating clear technical communication and accountability.

ASSESSMENT

How the course is graded



40%

Final exam

Closed book · Topics 6–12

30%

Labs and capstone

Lab reports and the capstone project

20%

Midterm exam

Closed book · Topics 1–5

10%

Quizzes

Short quizzes and class participation

Assessment methods and outcomes

Method	What it covers	Weight	Outcomes
Quizzes and participation	Short quizzes and active class participation, covering all topics	10%	CLO1, CLO2
Lab assignments and capstone	Weekly lab reports and the capstone project, with documentation and presentation	30%	CLO1, CLO2, CLO3
Midterm exam	Closed book. 20 multiple-choice, 10 short-answer and 1 AWS lab exercise. Topics 1–5	20%	CLO1
Final exam	Closed book. Multiple-choice, short-answer and problem-solving questions. Topics 6–12	40%	CLO2

Weight by learning outcome: CLO1 30% · CLO2 55% · CLO3 15%

Course policy

< 30

Overall score below 30 out of 100 means a re-sit of the exam.

> 50%

Absence without permission over 50% of the total course hours is an automatic fail.

3 weeks

No weekly project update for more than 3 weeks means zero for the poster presentation.

Student responsibilities

1. Create and maintain an AWS Academy Learner Lab account (quota 50 USD per account)
2. Monitor AWS usage to stay within the quota limit
3. Complete weekly lab assignments and submit lab reports on time
4. Work consistently on the capstone project throughout the semester
5. Participate actively in class discussions and demonstrations
6. Document all AWS resources created and clean them up after labs
7. Back up project code and configurations regularly
8. Attend all lectures and lab sessions, and adhere to academic integrity policies

Course schedule · Weeks 1–8

Week	Topic	Lab or project
1	Introduction to cloud computing and its evolution	—
2	Deployment and service models: IaaS, PaaS, SaaS	—
3	Cloud infrastructure, EC2 and virtualization	Lab 1: AWS account and EC2
4	Cloud scaling and load balancing	Lab 2: Auto scaling and ELB
5	Cloud storage: S3, EBS, EFS and data management	Lab 3: S3 storage
6	Cloud networking (VPC) and content delivery	Lab 4: VPC and CloudFront
7	Databases in the cloud: RDS and DynamoDB	Lab 5: RDS and DynamoDB
8	Open week, no topic listed in the specification	—

Course schedule · Weeks 9–16

Week	Topic	Lab or project
9	Cloud-native applications and microservices	Lab 6: Containers (ECS/EKS)
10	Security, IAM and compliance in AWS	Lab 7: IAM and security
11	Cloud cost management and economics	Lab 8: Cost monitoring
12	DevOps, CI/CD pipelines and Infrastructure as Code	Capstone: planning
13	AI/ML services in the cloud	Capstone: development
14	Capstone project work	Capstone: implementation
15	Review of the whole course	—
16	Final examination and project submission	—

Tools and references

SOFTWARE

Tools

AWS Console, AWS CLI and AWS Academy Learner Lab

VS Code

Git

Docker

Terraform

Postman and a web browser

READING

Textbooks

Required: AWS Certified Solutions Architect Official Study Guide, 3rd edition. Baron, Baz, Bixler, Gaut et al. Sybex, 2020.

Recommended: Cloud Computing: Concepts, Technology and Architecture (Erl, Puttini, Mahmood; Prentice Hall, 2013) and Architecting the Cloud (Kavis; Wiley, 2014).

LEARNING

Online resources

AWS Documentation

AWS Well-Architected Framework

AWS Whitepapers

AWS Training and Certification

AWS Labs and hands-on tutorials

Labs

AWS Academy Cloud Foundation

50% coupon toward the AWS Certified Cloud Practitioner exam.

AWS Educate

Getting started with compute, networking, security, storage, databases and serverless.

AWS Academy Learner Lab

Hands-on lab account for the weekly exercises, with a 50 USD quota per account.

Bonus for AWS Certified students

Students who earn an AWS certification receive bonus credit.

Week 1 learning objectives

1. Define cloud computing
2. Describe the five essential characteristics of cloud computing
3. Summarize the history and evolution of cloud computing
4. List the benefits of cloud computing
5. Enumerate the challenges of cloud computing

THIS WEEK

Topic 1

Introduction to cloud computing: concepts and evolution. Assessed by class participation and a quiz (CLO1).

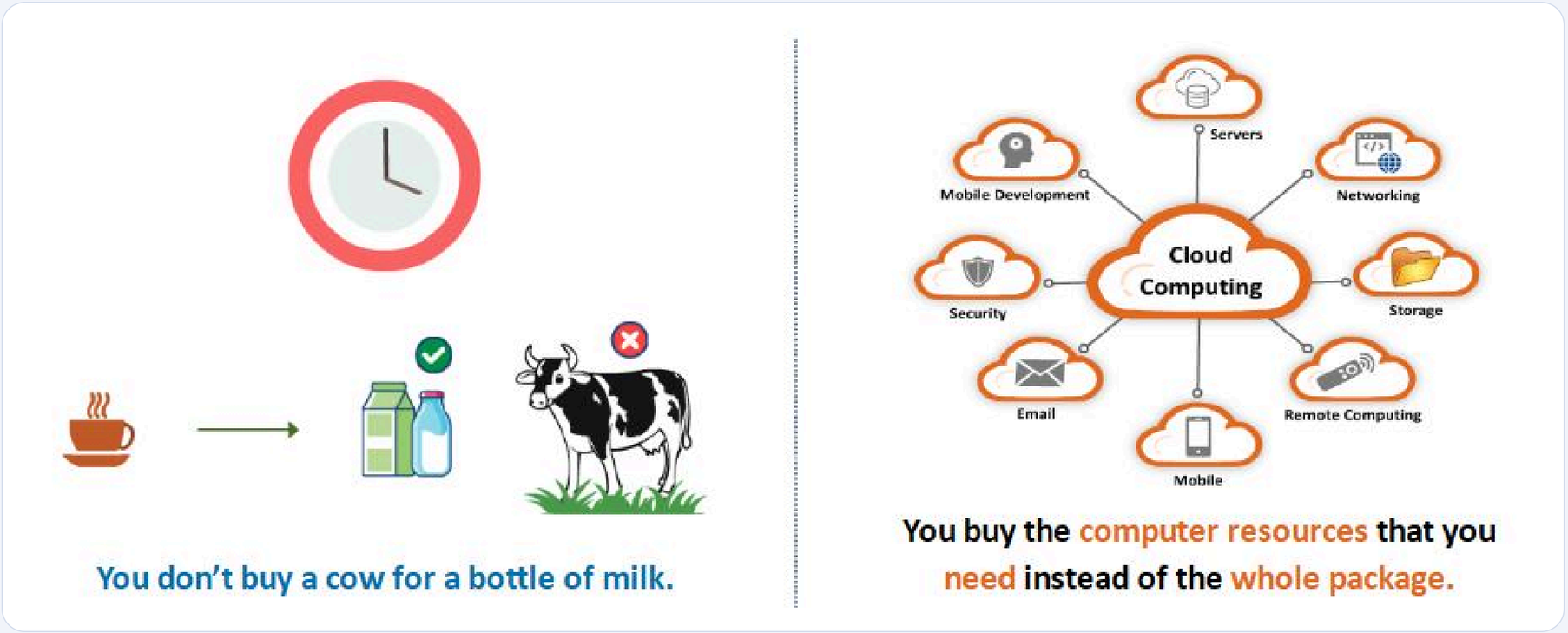
DISCUSSION

What is the cloud?

Join at slido.com with the event code
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[Slido logo]

A cloud computing analogy

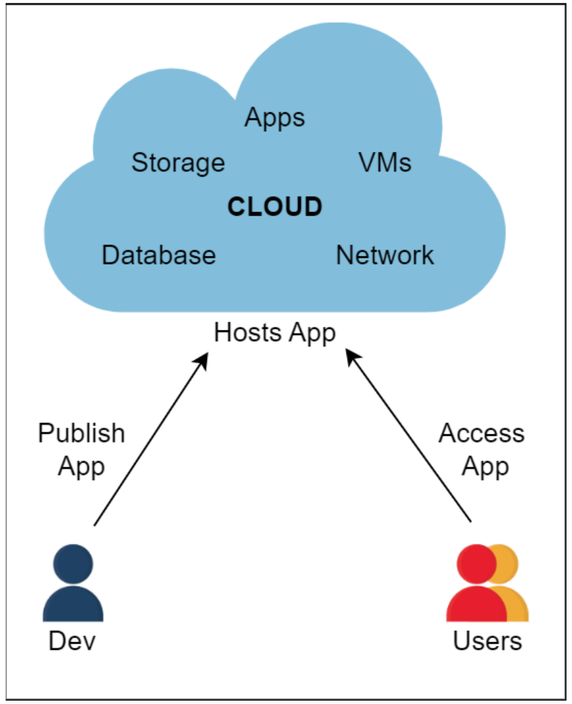


You buy the computing resources you need, not the whole package.

What is cloud computing?

The on-demand delivery of computing services *over the internet*, “the cloud”.

Servers	**Storage**	**Databases**
Networking	**Software**	**Analytics**



Three definitions

Wikipedia

Shared pools of configurable system resources and higher-level services, rapidly provisioned with minimal management effort.

AWS

On-demand delivery of compute power, database storage, applications and other IT resources over the internet, with pay-as-you-go pricing.

NIST

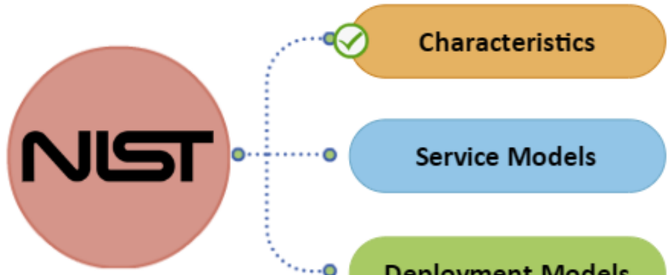
Convenient, on-demand network access to a shared pool of configurable computing resources, rapidly provisioned and released with minimal effort.

In this course the NIST definition is the standard.

A cloud is just a bunch of computers in a datacenter.

It is a real physical place, reached over the internet, where a service is performed for you or your stuff is stored. Explain the Cloud like I'm 10, Todd Hoff.

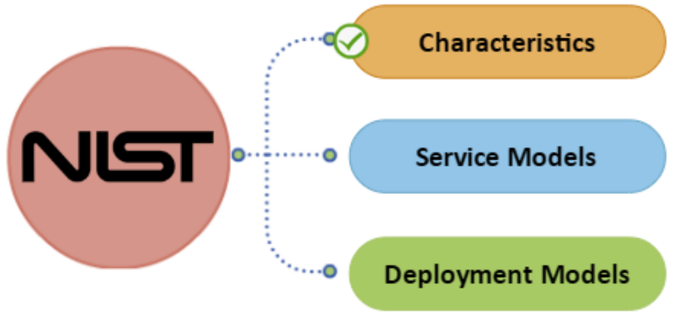
NIST key takeaways



Characteristics: the five essentials of cloud computing

1. On-demand self-service
2. Broad network access
3. Resource pooling
4. Rapid elasticity
5. Measured service

NIST service models

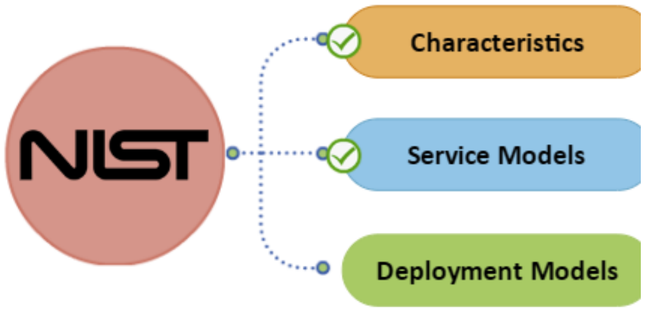


Software as a Service (SaaS)

The cloud provider's application is hosted on a cloud infrastructure and used by customers across the world.

For example, Gmail is a cloud-hosted application run by Google.

NIST service models



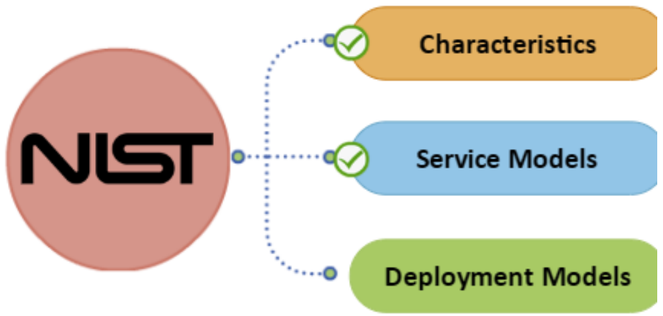
Platform as a Service (PaaS)

Applications deployed on the cloud infrastructure are offered by the cloud service provider. For example, deploying applications.

Infrastructure as a Service (IaaS)

The provisioning of fundamental computing resources for application deployment. For example, creating virtual machines on Amazon Web Services or Microsoft Azure.

NIST deployment models



Public cloud

Infrastructure for the general public. Government services used by the public are public cloud.

Private cloud

Infrastructure solely for a single organization, such as applications used in schools or offices.

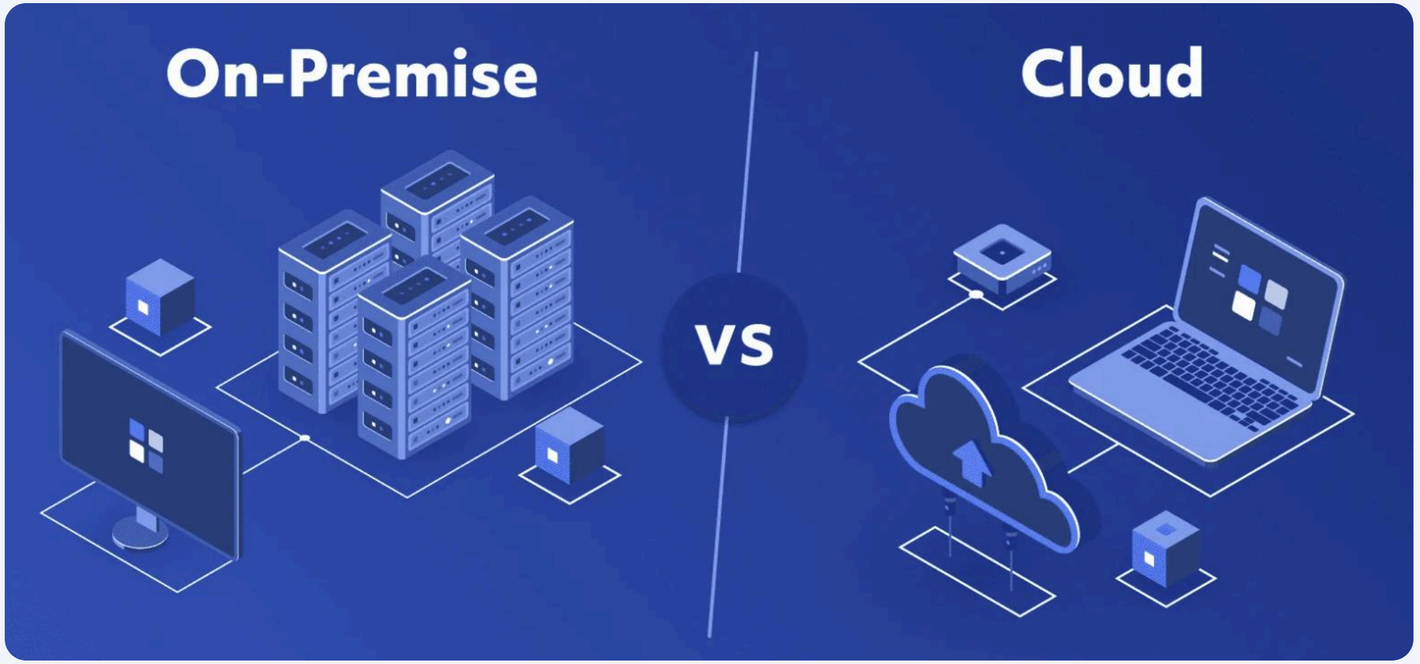
Community cloud

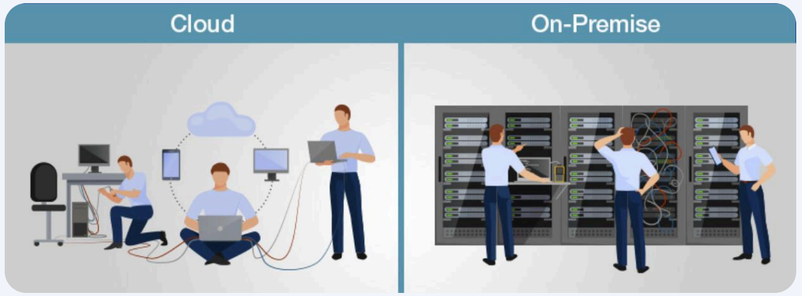
Infrastructure for a community with similar concerns. For example, Facebook.

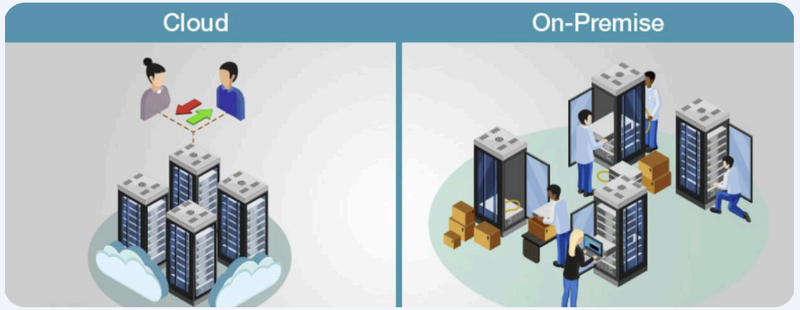
Hybrid cloud

A combination of two or more clouds, used by an organization or academic institution.

On-premises vs cloud: why businesses shift



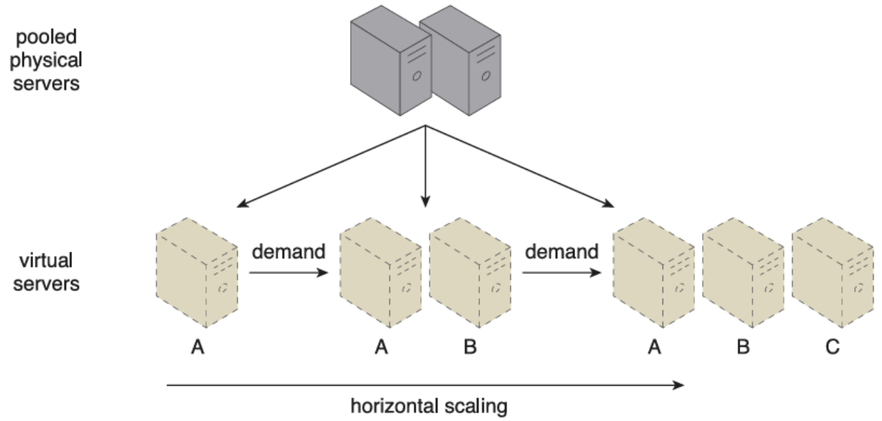




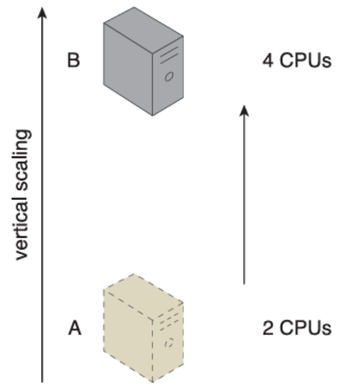
Why do businesses shift from the traditional on-premises model, hosting on local servers, to the cloud computing model?

Flexibility and scalability

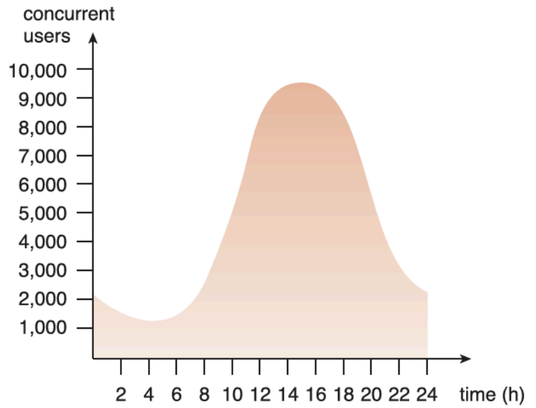
Quickly scale up or down based on demand.



Horizontal scaling: add virtual servers as demand grows



Vertical scaling: more CPUs per server



Concurrent users peak during the day

Cost efficiency

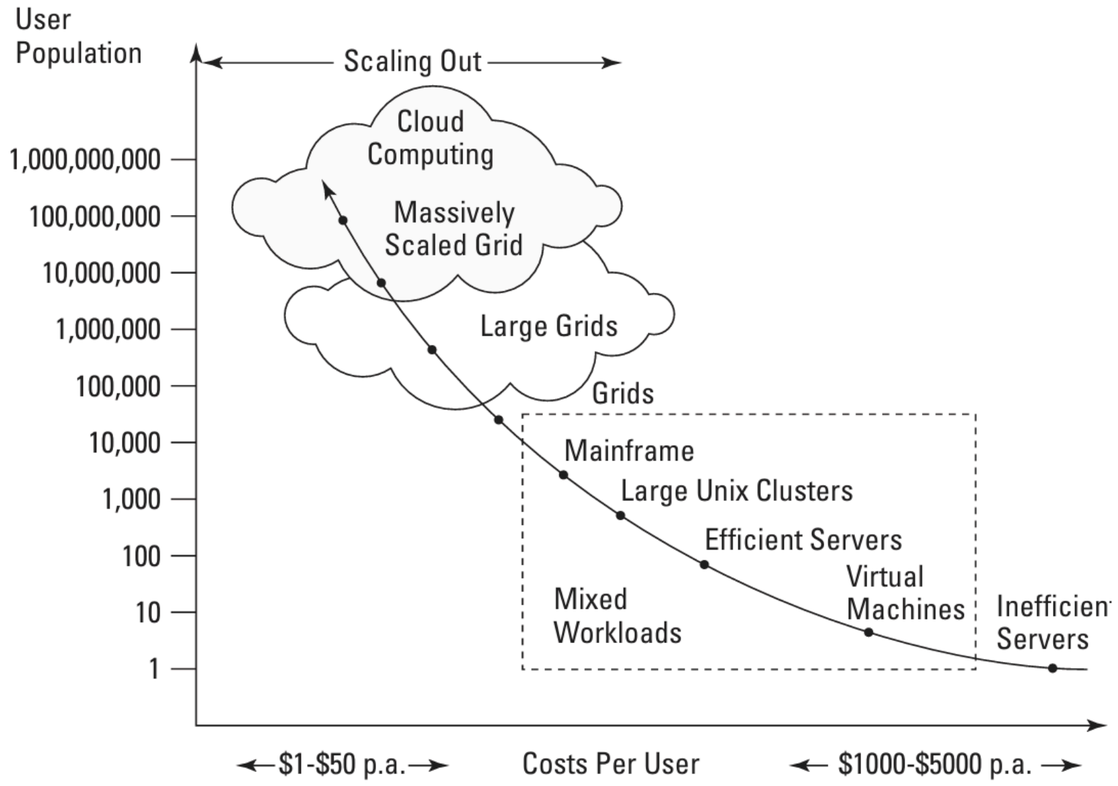
Eliminate the capital expense of buying hardware and running on-site datacenters. Pay for what you use with flexible pricing models.

$1,000–$5,000

per user per year with inefficient servers

$1–$50

per user per year at cloud scale



Performance and speed

Providers use the *latest, fast, efficient hardware* and global datacenters to reduce latency.

Regions

Datacenters are grouped into regions around the world.

Availability zones

Each region holds several zones, so applications can run close to their users.



Availability

Users can customize their applications using cloud services from *anywhere*.

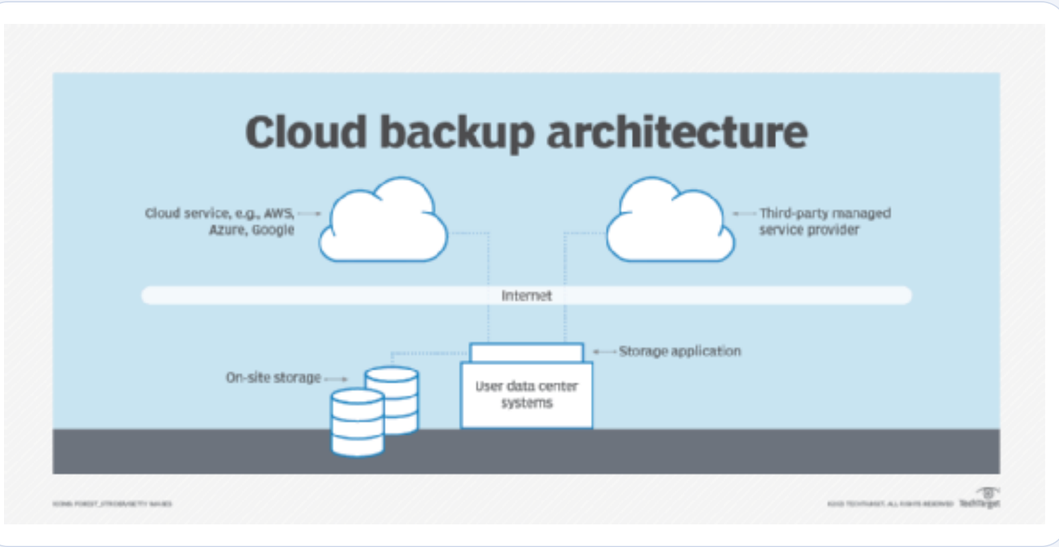
Cloud services reach laptops, tablets, phones and watches over the internet.



Security



Datacenters are physically secured and run by the provider.



Cloud backup: on-site data is copied to a cloud service or a managed provider over the internet.

Why businesses move to the cloud

Flexibility and scalability

Scale up or down quickly as demand changes.

Cost efficiency

No upfront hardware or on-site datacenter. Pay for what you use.

Performance and speed

Modern hardware and global datacenters reduce latency.

Availability

Use and customize your applications from anywhere.

On-premises vs cloud

	On-premises	Cloud
Upfront cost	Buy hardware, build a datacenter	No capital expense, pay as you go
Scaling	Buy and install more servers	Scale up or out on demand
Maintenance	Your own team runs it	Provider runs the hardware
Access	Local server	Anywhere over the internet

On-premises vs cloud comparison

Five essential characteristics

01

On-Demand Self-Service

Provision and release resources yourself, with no human help from the provider.

02

Broad Network Access

Reach services over the network from laptops, phones and tablets.

03

Resource Pooling

Shared physical and virtual resources serve many tenants, assigned by demand.

04

Rapid Elasticity

Scale out or in quickly, even automatically, so capacity feels unlimited.

05

Measured Service

Usage is metered and reported, so you pay for what you use.

Source: NIST

Why know these characteristics?

Self-service

Access resources anytime without waiting.

Anywhere

Use services from any location.

Shared

Efficient and cost-effective for many users.

Scaling

Adjust resources to your needs.

Pay for use

Only pay for what you actually use.

A shared language

NIST's definition is the standard reference, so providers, customers and engineers describe the cloud the same way.

Real cloud or not

If a service lacks self-service or metering, it is only hosting. The five traits are a checklist for spotting real cloud.

Better decisions

Each trait ties to a benefit and a cost, so you can choose, size and budget cloud services with confidence.

History of cloud computing

1960s

Mainframe timesharing: users rent a time slot to run calculations.

1970s

Virtual machines emerge and are rented out with private networks.

1990s

The web arrives, servers are virtualized and computing becomes a utility.

2006

Amazon launches AWS, lending its spare computing capacity to the market.

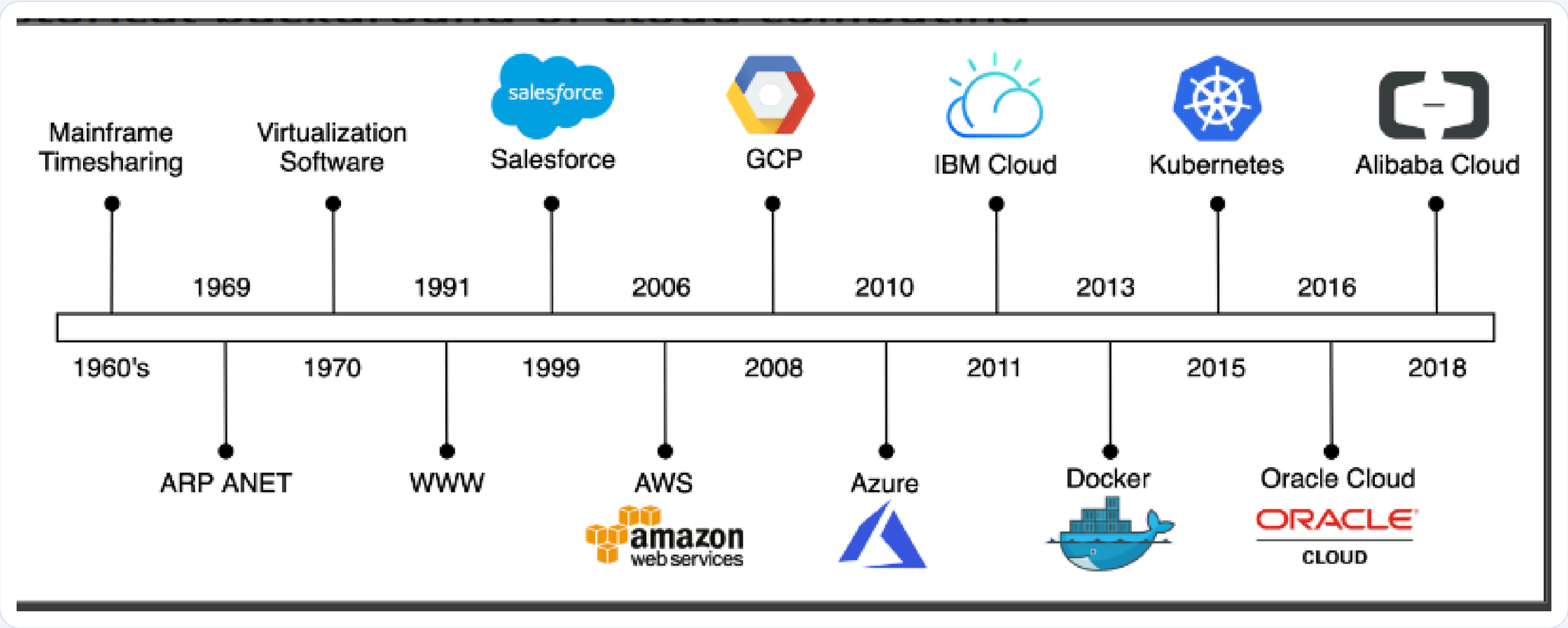
2008–10

Google Cloud and Microsoft Azure join the race.

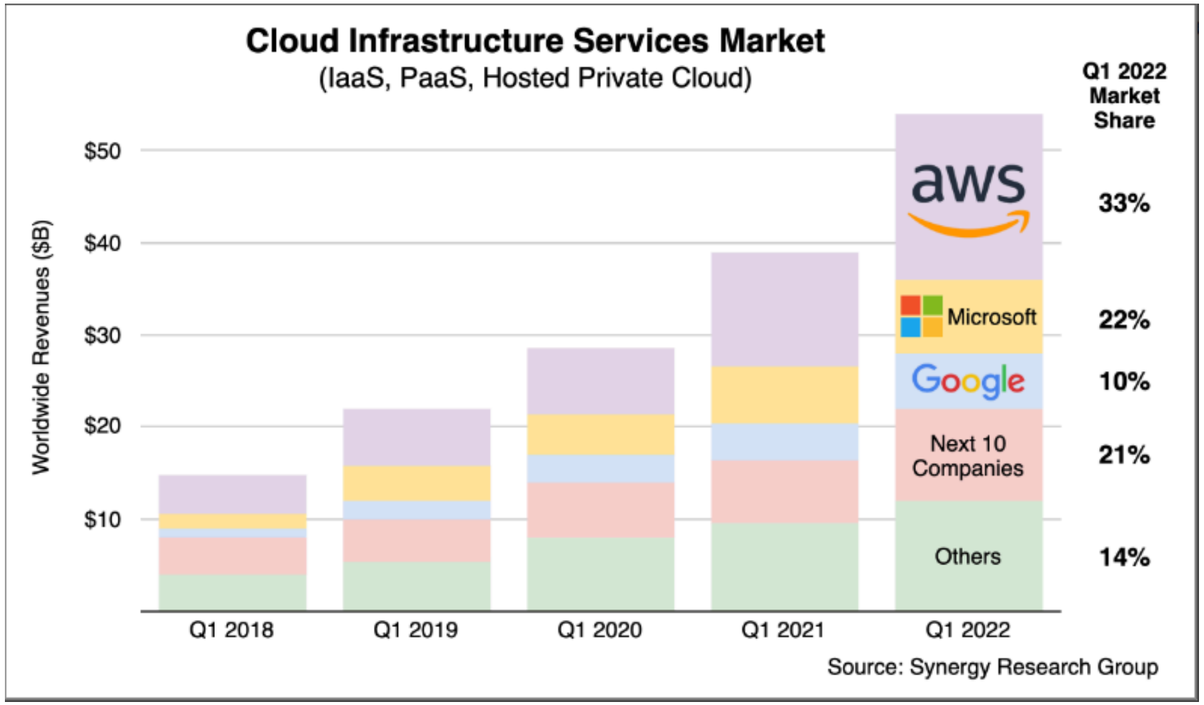
2013+

Docker and Kubernetes make containers the cloud-native standard.

Cloud computing timeline



Major providers of cloud services



33% AWS

22% Microsoft Azure

10% Google Cloud

Source: Synergy Research Group, Q1 2022 market share

First-tier and second-tier providers



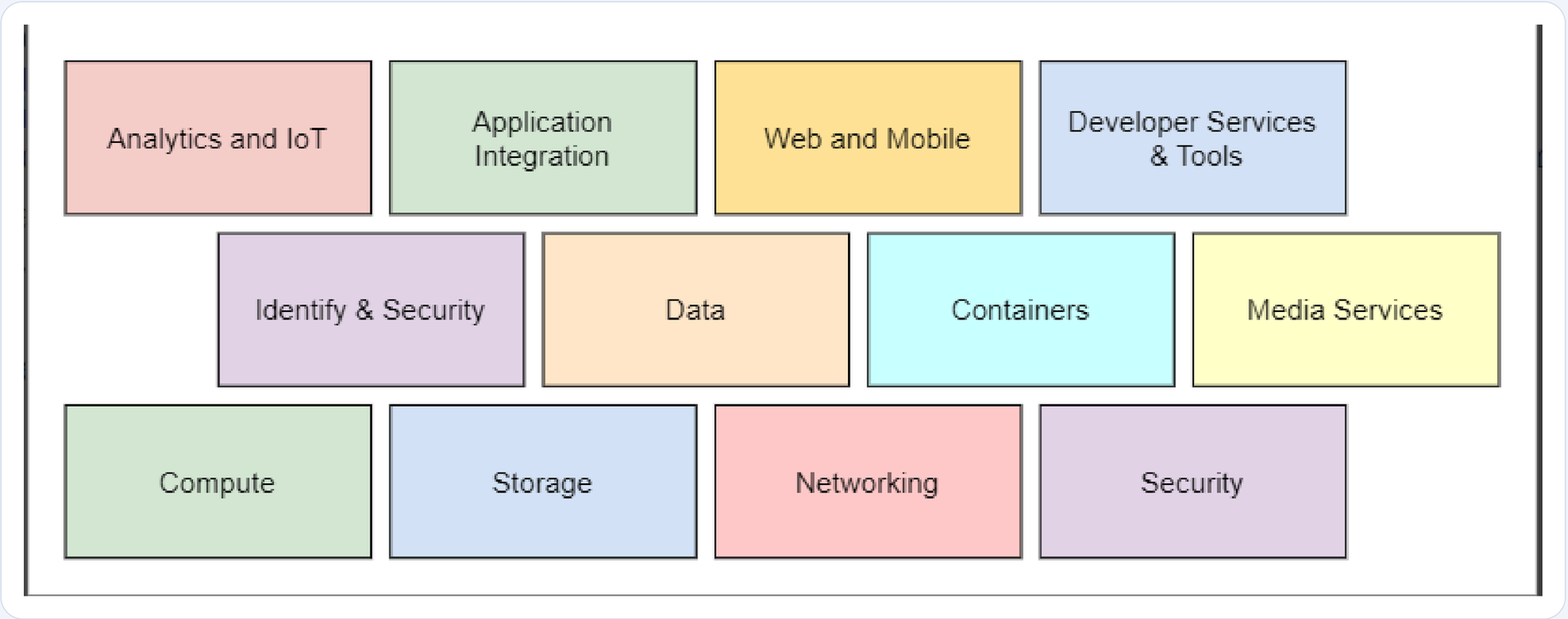
First tier

AWS · Microsoft Azure · Google Cloud Platform

Second tier

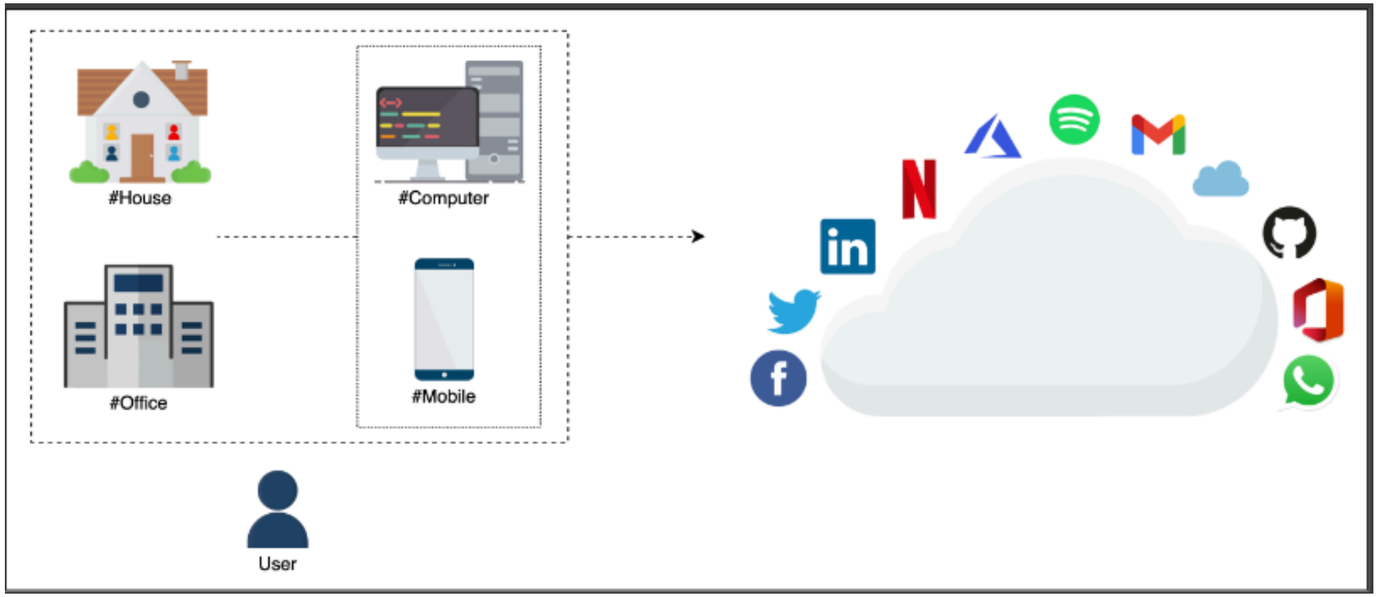
IBM Cloud · Alibaba Cloud · Oracle Cloud · Tencent Cloud · Salesforce · SAP

Cloud services



There are hundreds of services, and they are similar across platforms: an AWS service usually has a match in Microsoft Azure.

Where the cloud is used



Most of the services we use every day run on cloud platforms: Netflix, Amazon Prime, Gmail, Spotify, Uber, WhatsApp, iCloud and GitHub.

These billion-dollar companies do not run their own datacenters. They focus on the customer experience and leave running and maintaining services to cloud vendors.

Largest companies on cloud platforms

Some of the Largest Companies Using Cloud Platforms

Company	Number of Active Monthly Users	Cloud Platform
Netflix	200+ million	AWS
Spotify	433 million	GCP
Twitter	450+ million	GCP
Instagram	2+ billion	AWS
Adobe (Creative Cloud)	30+ million	Microsoft Azure / AWS

AWS

Netflix and Instagram

Google Cloud

Spotify and Twitter

Azure and AWS

Adobe Creative Cloud

Some use cases of cloud

Software

Delivery of software and services.

Streaming

Audio and video.

Cloud-native

Cloud-native applications.

Backup

Store, back up and recover data.

AI

Data analysis, machine learning and AI workloads.

DISCUSSION

What do you think could be the cons of cloud?

Join at slido.com with the event code **[code]**

Challenges of cloud computing



Cost

Running in the cloud is not automatically cheaper. Choosing the right service for the use case is part of the design.



Security

Protecting sensitive data has been debated since the start, and data breaches remain a real concern.



Vendor lock-in

Switching providers can cost so much that customers stay even when they are unhappy.

Why should we use the cloud?

Developer or software engineer

Working knowledge of the cloud is almost essential; most organizations run, or are migrating, their workloads there.

IT professional or admin

These roles have shifted heavily toward the cloud, where most hardware and software is managed.

Executive

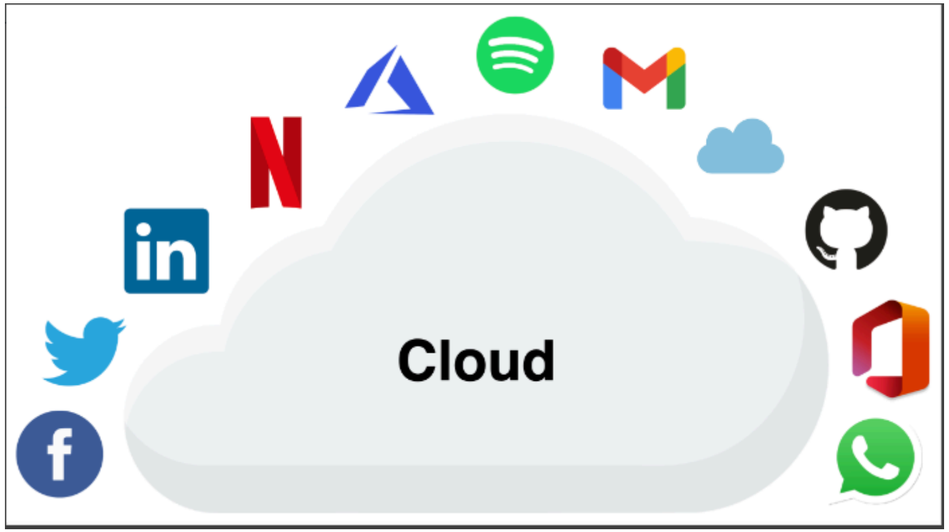
You are part of the digital transformation that almost every business is going through.

Entrepreneur

One person or a very small team can build a robust business on a global scale.

Consumer

Everyday apps such as Netflix, Gmail and WhatsApp run on the cloud.



Emerging technologies powered by cloud



IoT

Combating rhino poaching in South Africa.



AI

Digital experiences for millions of United States Tennis Association fans.



Blockchain

Food supply chain traceability that helps farmers reduce waste.



Analytics

Predictive maintenance for a city's infrastructure, by KONE.

Next session



01

Common cloud services



02

Cloud deployment models



03

Cloud service models